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teries that can be printed cheaply. Yes, printed! In the meantime, smaller batteries mean shorter battery life, which will need advancements in wireless power and charging, over and above what wireless power alliances, like Qi and A4WP, are supporting today.

"A lot of research is being done to solve the battery-power issue. Imprint Energy has come up with an ultra-thin and flexible battery that can power your wearable gadget, by making it less bulky and more designer friendly. They have used Zinc battery technology, called ZincPoly, which is stable and safer than conventional lithium based batteries. Screen printing is the technology used by them to create custom-shaped batteries for wearables," says Arvind Sanjeev, founder of DIYHacking.com and creator of the indigenous Google Glass clone.

He adds that transparent wearables is something that is gaining a lot of attention. Imagine wearing a transparent band/watch, wouldn't it be cool? The major hindrance towards making this a reality was batteries, which had to be transparent too. However, SLAC National Accelerator Laboratory has come out with a solution to this as well, by making transparent batteries using a transparent, slightly-rubbery compound known as poly-di-methyl-siloxane (PDMS).

T. Anand also explained about an alternative battery technology that sounds interesting. "A new alternative in the form of solid-state batteries is coming up. These batteries would be fabricated over silicon dye and would undergo almost the same process as that of chip manufacturing. These batteries are expected to

have very-high-density energy stored, just like a chip would contain millions of transistors."

Materials technology is now more significant than ever

When the game is about building products that are in constant contact with human skin, choice of material gains a new level of significance.

While selecting materials might seem purely cosmetic to some, there are examples where a wrong selection has resulted in a rash of recalls (notice the pun). Fitbit Force was one such product that had to be recalled due to complaints of irritation by customers. The cause was presumed to be either the nickel in the stainless-steel part of the device or adhesives used in the strap. This was so serious that it resulted in the initiation of an investigation by Consumer Product Safety Commission of the UK. Another example is the use of zinc in Ring gesture device by Logbar, which caused signal interference.

"Over US\$ 100 billion will be spent on materials for wearable electronics in the coming decade," reports Dr Peter Harrop and Dr Guillaume Chanin in their research report, *Wearable Technology Materials 2015-2025*.

While sports products like my Jawbone UP24 came with hypo-allergic medical-grade TPU rubber, premium products like Apple's iWatch are claimed to come with stainless steel, silver aluminium or 18-karat yellow and rose gold. Nike's FuelBand came built with a mix of thermoplastic elastomers, polypropylene, magnesium and stainless steel. Fitbit band was also marketed as having a

flexible, durable elastomer material without latex. Fitbit's clasp is made of surgical-grade stainless steel and contains traces of nickel.

Looking at the material space, it looks like luxury brands have their own game to play here. For instance, Mont Blanc (yes, the old upmarket pen company) has launched an e-strap for their line of luxury watches. Made from leather with a carbon-fibre texture, it holds an organic LED display in a carbon-coated steel housing.

In an example of taking it all the way, Austrian luxury brand Swarovski has teamed-up with Misfit to create a giant crystal, embedded with an activity tracker that can be matched with whatever pieces of jewellery you have from Swarovski. This could be a dangerous combination for your purse if you have a nerdy girlfriend, but it is still impressive from a purely technical standpoint.

Will society accept wearing and implanting electronics

When computers were first introduced, there was uproar on the jobs it would displace, the automations that would substitute humans at work and the emotion-less digitisation of life. Today, we have computers on our desks and laps, in our cars, homes and pockets. Wearables and implantables will probably have the same destiny. Dr Nagesh tells us that, under Obama Care, America is planning to use implantables for greater social benefit.

It seems apt to use the most-used dialogue of the cybernetic *Borg* here, and quote: "Resistance is futile." ●